

BELIMO

Steam Valve Sizing Tool

(Belimo Steam Valve Sizing Tool.xlsm)

USER MANUAL

Release 3.6

Prepared for Belimo customers, contractors, and internal staff
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1. Introduction

The Belimo Steam Valve Sizing Tool (the “tool,” file name Belimo Steam Valve Sizing Tool.xlsm) is an Excel-based calculator that helps you pick the right steam control valve for a heating coil, fin-tube radiator, heat exchanger, or other steam load. You tell it what the application is and what the steam conditions are, and it works out the valve size (Cv or Kv) you need, checks the flow for choked (critical) flow conditions, recommends a Belimo valve that passes both the capacity and the velocity check, and checks pipe velocities so you don't end up with a noisy or oversized installation.

This manual is written for two audiences at once: engineers and contractors sizing a valve for a real job, and Belimo staff who need to explain or troubleshoot the tool for a customer. Wherever the tool uses an engineering term (like “choked flow” or “critical pressure ratio”), this manual explains it in plain language first, then gives the technical detail for anyone who wants it.

Who this manual covers

Whether you are a contractor filling in numbers for a single job, or a Belimo application engineer walking a customer through the tool over the phone, every dropdown, button, and result on every sheet is documented here — in order, with screenshots.

1.1 What the tool does, in one paragraph

You enter the application type, whether the valve is on/off or modulating, the steam flow rate, the inlet and outlet pressures, and your elevation/location (for barometric pressure correction). The tool calculates the required valve coefficient (Cv or Kv), tells you whether the flow is choked (a special condition explained in Section 11), recommends the smallest Belimo steam valve that passes both the capacity and the outlet-velocity check, and — once you tell it your pipe sizes — checks that steam velocity through the valve and piping stays in a safe, quiet range.

1.2 What's in this manual

- Section 2 – Getting started: opening the file, enabling macros, and the About screen
- Section 3 – The Calculator sheet: every input, dropdown, and result, step by step.
- Section 4 – Feedback, release information, and changing the tool's language.
- Section 5 – The 1/3-2/3 split-range valve calculator.
- Section 6 – The Summary sheet, for logging multiple sizing results.
- Section 7 – The Tool Box, a set of extra unit-conversion and engineering calculators.
- Section 8 – The Steam Pipe Capacity reference table.
- Section 9 – The Valve database reference.
- Section 10 – The standalone Pipe & Valve Velocity checker.
- Section 11 – The engineering behind the tool, explained simply.
- Section 12 – Troubleshooting and getting help.

- Appendix A – Every dropdown in the tool, in one table, for quick reference.

2. Getting Started

2.1 Opening the file

Double-click Belimo Steam Valve Sizing Tool.xlsm to open it in Excel. Because the tool uses macros (VBA) to run its calculations, Excel will show a yellow “Security Warning” bar under the ribbon the first time you open it.

- Click “Enable Content” (or “Enable Macros”) on that yellow bar.
- If the file was downloaded from email or the internet, Windows may also ask you to confirm you trust it — this is normal for any macro-enabled workbook and is safe for this file.

Why macros are required

All of the sizing math, the valve auto-selection, the unit conversions, and the help tooltips are built with VBA (Excel's built-in programming language). If macros are disabled, the buttons and automatic calculations on the Calculator sheet will not work.

2.2 The About screen

Click the small (i) About button near the top of the Calculator sheet, next to the Close Tool button, to open the About screen. It shows the tool's name, its version and release date, an internal-use notice, the tool's disclaimer, and who maintains the tool. It also has three buttons – Check for Updates, Download Fresh Copy and Download User Manual (see Section 4.2). The disclaimer is there for information: the tool's results are meant to guide valve selection and should always be verified against the actual system design conditions.

2.3 Where you'll spend most of your time

Almost everything you need for day-to-day sizing lives on one sheet: Calculator. The other sheets (OneThirdTwoThird, Summary, Tools, SteamPipeSize, Valves, Velocity, and the language/update sheets) are all reference tools or specialized calculators that support the main Calculator sheet, and are covered in Sections 5–10. You switch between them using the sheet tabs along the bottom of the Excel window.

3. The Calculator Sheet — Step by Step

This is the heart of the tool. Fill in the highlighted input cells from top to bottom, and the results fill themselves in as you go. The screenshot below shows the sheet with no inputs entered yet, so you can see where everything is.

The screenshot shows the 'Belimo Steam Valve Sizing Tool' spreadsheet. It features a top navigation bar with 'App' and a formula bar. The main area is divided into several sections:

- APPLICATION DATA:** Includes dropdowns for 'Application [APP]' and 'On/Off or Modulating Control? [Ctrl]'.
- STEAM DATA:** Includes input fields for 'FLOWRATE [Q]' (0.00 LB/HR), 'Supply INLET Pressure [Pi]' (0 PSIG), 'OUTLET Pressure [Po]' (0 PSIG), and 'Atmospheric Pressure [AP]' (14.7 Default).
- Cv CALCULATOR RESULTS:** A table of calculated values including 'Converted Flowrate [Q]', 'Downstream Psig [DSP]', 'PSI Drop [h]', 'Inlet Pressure [Pi]', 'Outlet Pressure + AP [Poa]', 'Flow Coefficient Result [Cv]', 'Selected Valve Input [Cv]', 'Corrected Pressure Drop', 'Corrected Drop Ratio', 'Flow Regime', 'Inlet Absolute Pressure', 'Corrected Outlet Abs Pressure', 'Corrected Flow Output', 'Required [Cv]', 'Difference [Cv]', 'Margin [%]', 'Selected Valve Outlet Velocity', and 'Selected Valve Noise'.
- CORRECTED PRESSURE DROP [H] & [Q] BASED ON VALVE Cv:** A section for pressure drop calculations.

A diagram of a 'G6 Series Globe Valve' is also present, showing its internal components and flow paths.

Figure 1 – The Calculator sheet, blank / default state.

3.1 Application

Cell D6. This tells the tool what kind of steam load you're sizing the valve for. This mainly affects which help text and defaults are shown — the sizing math itself is the same steam-flow calculation regardless of application.

Field / Dropdown	Where it is	Choices
Application	Calculator sheet, cell D6	Coil, Fin Tube, Heat Exchanger, Radiator

3.2 Control (Ctrl)

Cell D8. Tells the tool whether the valve will be a simple open/closed (on/off) valve or a modulating (throttling, variable-position) valve. Modulating applications generally call for equal-percentage valves for good control across the full stroke, while on/off applications are less sensitive to valve characteristic.

Field / Dropdown	Where it is	Choices
Control	Calculator sheet, cell D8	On/Off, Modulating

3.3 Flow Rate and Flow Unit

Cell D12 (value) with its unit dropdown. Enter the design steam flow rate for the load, then choose the unit it's expressed in. The tool converts everything internally to lb/hr before running the sizing calculation, so you can enter the number in whichever unit you have on hand — a boiler rating in BTU/HR, a coil rating in EDR, or a straightforward lb/hr or kg/h flow.

Field / Dropdown	Where it is	Choices
Flow Unit	Calculator sheet, cell D12 (unit selector)	BTU/HR, EDR, LB/HR, MBH, KG/h

3.4 Inlet Pressure (PI) and Outlet Pressure (PO)

Cells D13 and D14 (with unit selectors at D13/T-column and D14/P-column). Enter the steam pressure just upstream of the valve (inlet, PI) and just downstream of the valve (outlet, PO). Both can be entered in PSIG (pounds per square inch, gauge) or BAR — pick whichever matches your job data.

Field / Dropdown	Where it is	Choices
Inlet Pressure Unit (PI)	Calculator sheet, cell D13	PSIG, BAR
Outlet Pressure Unit (PO)	Calculator sheet, cell D14	PSIG, BAR

The difference between these two pressures (the pressure drop across the valve) is what drives steam through the valve — it's the single biggest factor in how large a valve you need. See Section 11 for why the ratio between PO and PI (not just the difference) also matters.

3.5 Location / Barometric Pressure (AP)

Cell D15. Because gauge pressure readings (PSIG) are relative to local atmospheric pressure, and atmospheric pressure changes with elevation, the tool needs to know roughly where the installation is so it can correct PSIG readings to absolute pressure correctly. Pick your U.S. state (or “Manual / Other” to type your own value, or “Default” to use standard sea-level atmospheric pressure) from the dropdown.

Field / Dropdown	Where it is	Choices
Location (Barometric Pressure)	Calculator sheet, cell D15	Default, Manual / Other, plus all 50 U.S. states and additional Canadian provinces (74 entries total — see Appendix A)
In plain English Think of this dropdown as telling the tool “how high up am I,” because a gauge pressure reading of 0 PSIG means something slightly different in Denver than it does in Boston. Picking your state gets this right		

automatically; “Manual / Other” lets you type an exact local barometric pressure if you have one.

3.6 Coefficient Type (Cv or Kv)

Cell D28. Choose whether you want the required valve size expressed as Cv (the U.S./Imperial flow coefficient) or Kv (the metric flow coefficient). The tool calculates and displays the result in whichever one you pick.

Field / Dropdown	Where it is	Choices
Coefficient Type	Calculator sheet, cell D28	Cv, Kv
Cv or Kv everywhere The choice applies to every result: the required value, the Valve Selection list, the Recommended Valve panel and the 1/3-2/3 split-range sheet (Section 5). The tool always works in Cv internally, so switching changes only the numbers shown, never a sizing decision.		

3.7 Reading the results

Fill in Application, Control, Flow, Inlet Pressure and Outlet Pressure, then click CALCULATE; the results fill in below and on the right. After changing an input, click CALCULATE again. The screenshot below shows a completed example: a Coil application, Modulating control, 2,500 lb/hr flow, 60 PSIG inlet, 40 PSIG outlet.

AP 14.7

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Belimo Steam Valve Sizing Tool

SHOW / HIDE STEPS CLOSE TOOL

APPLICATION DATA

Application [APP] Coil

On/Off or Modulating Control? [Ctrl] Modulating

STEAM DATA

FLOWRATE [Q] 2500.00 LB/HR

Supply INLET Pressure [PI] 60 PSIG

OUTLET Pressure [PO] 40 PSIG

Atmospheric Pressure [AP] 14.7

RESET CALCULATE DISPLAY CV'S ADD TO SUMMARY

G6 Series Globe Valve

Cv CALCULATOR RESULTS

Converted Flowrate [Q] 2500.00 LB/HR

Downstream Psig [DSP] 40.00 psig (gauge)

PSI Drop [h] 20.00 psig (gauge)

Inlet Pressure [PI] 60.00 psig (gauge)

Outlet Pressure + AP [Poa] 54.70 psia (absolute)

Flow Coefficient Result [Cv] 23.04 Cv

Standard Calculation

$$Cv = Q / (3.28 * P1_abs * SQRT(((P1-P2)/P1)*(P2/P1)))$$

$$v = 2500.0 / (3.28 * 74.700 * SQRT(((74.700 - 54.700)/74.700)*(54.700/74.700)))$$

Closest Cv (capacity only): 24.000 - B2100VB-024 (1-in)

CORRECTED PRESSURE DROP [H] & [Q] BASED ON VALVE CV

Selected Valve Input [Cv] 24 Cv

Corrected Pressure Drop 17.69 psi

Corrected Drop Ratio (ΔP_corr/P1) 23.68 %

Flow Regime Subcritical

Inlet Absolute Pressure (P1_abs) 74.70 psia

Corrected Outlet Abs Pressure (P2_corr_abs) 57.01 psia

Corrected Flow Output 2,603.72 lb/hr

Required [Cv] 23.04 Cv

Difference [Cv] 0.96 Cv

Margin [%] 4.15 %

Selected Valve Outlet Velocity 1043.9 ft/s

Selected Valve Noise 67.4 dBA

Valve Selection B2100VB-024 (1-in) - Cv 24.0

Figure 2 – A worked example, fully filled in with results calculated.

In this example the tool calculated a required Cv of 23.04 and put the closest catalog valve, the B2100VB-024 (Cv 24.0), in the Valve Selection box. That is a capacity-only match: Section 3.11 explains why the Recommended Valve further down is a different one.

3.7.1 Manually overriding the selected valve

If you want to size a specific valve model yourself instead of using the tool's automatic pick, use the valve selection dropdown (cell J23) to choose from the full Belimo steam valve product line directly.

Field / Dropdown	Where it is	Choices
Selected Valve (manual override)	Calculator sheet, cell J23	-Select-, plus all Belimo steam valve models with size and Cv shown, e.g. "B2050VS-01 (0.5-in) - Cv 1.0" ... "B2150VS-055 (1.5-in) - Cv 55.0" and so on (60 entries — see the Valves sheet, Section 9)

The tool checks the pressure rating of whatever valve you pick against the job's pressure conditions. If you select a valve that isn't rated for the pressure you've entered, a message box will warn you and ask you to choose a different model — this guards against accidentally specifying an undersized-pressure-class valve.

3.8 Recommended Valve & Pipe Size / Pipe & Valve Velocity Check

Scroll down on the Calculator sheet and you'll find two more result panels that appear once the main sizing is complete: a Recommended Valve & Pipe Size summary, and a Pipe & Valve Velocity Check section where you tell the tool your actual pipe sizes so it can verify velocity.

AP	A	B	C	D	E	F	G	H	I	J	K	L	M	N
31		$y = 2500.0 / (3.28 * 74.700 * \text{SQRT}(((74.700 - 54.700)/74.700)^2 + (54.700/74.700)^2))$					Difference [Cv]	0.96	Cv					
32		Closest Cv (capacity only): 24.000 - B2100VB-024 (1-in)					Margin [%]	4.15	%					
33							Selected Valve Outlet Velocity	1043.9	ft/s					
34							Selected Valve Noise	67.4	dBA					
35														
36														
37			Flow is subcritical: P2_abs / P1_abs = 0.732, which is greater than 0.58											
38														
39														
40														
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50														

Figure 3 – Recommended Valve & Pipe Size, and the Pipe & Valve Velocity Check panel.

3.8.1 Target Velocity

A small dropdown lets you choose the target design velocity (in ft/s) the tool should check pipe sizing against.

Field / Dropdown	Where it is	Choices
Target Velocity (ft/s)	Calculator sheet, cell C41	50, 80, 120

3.8.2 Velocity Check dropdowns

Once you've selected an actual Valve Type and entered your Supply Pipe Size, Outlet Pipe Size, and Valve Size, click Calculate Velocity to check that steam speed through each is within a safe, quiet range.

Field / Dropdown	Where it is	Choices
Valve Type	Calculator sheet, cell H42	Butterfly, Globe, HTCCV, V-Ball, VS, VSS
Steam Type	Calculator sheet, cell H43	Dry Saturated, Wet Saturated, Superheated
Supply / Outlet Pipe Size, Valve Size	Calculator sheet, cells H44–H46	-Select-, 0.5, 0.75, 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12 (inches)

If the calculated velocity is too high for the pipe or valve size you've chosen, the tool flags it with a caution message and (as shown in the earlier calculator example) tells you the recommended minimum pipe size instead, so you can correct the selection before it goes on a drawing.

3.9 Simple / Detailed view toggle

Field / Dropdown	Where it is	Choices
View (Simple / Detailed)	Calculator sheet, cell C50	Simple, Detailed

“Simple” hides the intermediate calculation steps and shows only the inputs and final results — good for a quick job or a customer-facing screen share. “Detailed” shows the full calculation trail (corrected pressure drop, choked-flow check, noise estimate, etc.) so an engineer can verify exactly how the tool arrived at its answer.

3.10 On-screen help: the “?” icons and guided walkthrough

Small “?” icons appear next to several inputs (Flow Rate, Inlet Pressure, Outlet Pressure, Barometric Pressure, and the velocity-check fields) and, since release 3.3, next to the key results (Section 3.11). Clicking one shows a short plain-language explanation of that item, right on the sheet, so help is always one click away without leaving the Calculator.

There is also a guided, numbered walkthrough (Steps 1 through 7) that can be shown or hidden as a set of on-sheet callouts pointing at each input in order, which is useful the first time you (or a customer) work through the tool, or when training a new team member.

[Reporting a problem](#)

If the tool ever shows an unexpected error, note what you were doing and what the message said, then use the Report a Bug or Give Feedback button on the Calculator sheet (Section 4.1) to report it. The more specific you are, the faster it can be tracked down.

3.11 Reading the results: the “?” icons on the results side

These icons open a plain-language message in the language you have selected. Three different valves can appear: the Closest Cv (capacity only), the Selected valve in the Valve Selection box, and the Recommended Valve (capacity and velocity) – only the last one is the answer. The 1/3-2/3 sheet has one more, beside its Recommendation (Section 5).

Icon next to...	What it tells you
Flow Coefficient Result [Cv]	The Cv (or Kv) needed to pass your flow at the pressures you entered – an opening size, not a valve position or a catalog valve. Choked flow is calculated at the critical pressure drop.
Closest Cv (capacity only)	The catalog valve nearest the required Cv, by capacity alone (no velocity check). It fills the Valve Selection box as a starting point; it is not the recommendation. DISPLAY CV’S lists each valve type’s closest rated Cv below and above required – also capacity only.
Above the Valve Selection box	The reverse question: what would the valve in the Valve Selection box (J23) do, fully open, at the same flow and inlet pressure? Not your operating conditions – a larger valve simply throttles.
Flow Regime	Subcritical, or choked (outlet at or below about 58% of the inlet, absolute). Choked flow depends on the inlet pressure only; expect more noise and erosion.
Required / Difference / Margin	Difference = selected Cv minus required Cv; Margin = that as a % of the required Cv. A large margin is not a fault (the next size down may fail the velocity check); a negative one means the valve is too small.
Selected Valve Outlet Velocity / Noise	The Selected valve’s own outlet velocity (ft/s; m/s for bar inputs) and noise (dBA, IEC 60534-8-3). Red: over the Steam Type limit, or 85 dBA and up – the icon explains the effect on noise, erosion and control.
Capacity check	PASS (green): the selected valve’s Cv covers the required Cv. FAIL (red): too small. PASS (orange): over 10% spare – check low-load control on the 1/3-2/3 sheet. The velocity line below it is the other half of the answer.
Target Velocity	Only sizes the inlet pipe (the outlet pipe is one size up); it does not change the required Cv or the valve’s velocity limit.
Valve Size & Cv	The Recommended Valve: the smallest valve rated for your inlet pressure with at least the required Cv whose outlet velocity at design flow (nominal outlet size, the Spirax Sarco basis) is within the Steam Type limit – wet 40 m/s, dry 150 m/s, superheated Mach 0.5. None over 3 times the required Cv: the panel then shows “None suitable – review with TS” and the note says why.
Valve Travel %	How far open the Recommended Valve would be at design flow – from Belimo’s measured curves for the G2 globes, otherwise estimated from the ideal curve (basis shown, e.g. “Equal %, R=50”). A guide, not a commissioning setpoint.

What the worked example shows

In Figure 2 the job needs a Cv of 23.04. The closest valve (B2100VB-024, Cv 24.0) passes the capacity check but fails the velocity check – 318 m/s (1,044 ft/s, in red on the right) against 150 m/s – so the Recommended Valve panel (Figure 3) picks the G240S-N (1.5-in, Cv 28.0), the smallest valve that passes both, at about 82% travel. Minimum load is then the question: see the 1/3-2/3 sheet (Section 5). A much larger drop – 60 to 10 psig at the same flow – would need a valve over 3 times the required Cv; the panel then shows “None suitable – review with TS”.

4. Feedback, Release Information & Language

Scroll to the bottom of the Calculator sheet and you'll find three more useful sections: a feedback link, the tool's version/update information, and the language selector.

The screenshot shows the bottom portion of a spreadsheet interface. At the top, a red warning message states: "Critical pressure drop reached. Critical downstream pressure limit = 28.63 psig". Below this, a section titled "SUGGESTED VALVE & PIPE SIZE" contains input fields for Target Velocity (ft/s) set to 80, Valve Size & Cv (G240S-N (1.5-in) - Cv 28.0), Valve Travel % (98% (Equal %, R=50)), Calculated Valve Noise (dBA) (OK), Valve Velocity (OK), Inlet Pipe Size (4-in Sch. 40 (min. for <=80 ft/s)), and Outlet Pipe Size (5-in Sch. 40 (1 size up from inlet)). To the right, a "PIPE & VALVE VELOCITY CHECK" section includes dropdowns for Valve Type (-Select-), Steam Type (Dry Saturated), Supply Pipe Size (-Select- inch), Outlet Pipe Size (-Select- inch), and Valve Size (-Select- inch), along with a "Calculate Velocity" button. Below these is a "Result:" field. The "FEEDBACK" section features a "Report a Bug or Give Feedback" button. The "RELEASE INFORMATION" section displays: Release Version 3.1, Release Date 9/23/2026, Update Mode 1, and Version Check Status Success. The "Language" section shows flags for US, UK, DE, and FR, with a "Remember my language" checkbox.

Figure 4 – Feedback, Release Information, and Language sections at the bottom of the Calculator sheet.

4.1 Feedback

Click the Report a Bug or Give Feedback button in the Feedback section to open Belimo's bug and feedback form in your web browser. Use it for anything from a typo to a feature request – it's the fastest way to get a change considered for the next release. If you are reporting a problem, tell us what you were doing when it happened (the values you had entered and any message the tool showed); the more specific you are, the faster it can be tracked down.

4.2 Release Information

This box shows the current Release Version and Release Date of your copy of the tool, along with the Update Mode and Version Check Status. The tool checks a Belimo-hosted update source each time it opens; "Success" (shown in green) means it confirmed you're on a supported version. If a newer version is available, you'll be offered the chance to view what's new and, if you choose to install it, pick where to save the new copy yourself through a standard Windows Save As dialog – just like saving any other file. Your current copy is never touched or deleted, so there's nothing to lose by trying it.

You don't have to wait for the automatic check. The (i) button near the top of the Calculator sheet, next to the Close Tool button, opens the About screen (Section 2.2). Check for Updates runs the same check on demand. Download Fresh Copy re-downloads the current published version right

away – useful if you suspect your copy is corrupted or missing something, even if you're already on the latest release. Download User Manual saves the current published copy of this manual to a folder you choose; if no manual has been published yet, the tool tells you so.

4.3 Language

Click one of the four flag icons to switch the entire tool's labels, buttons, and messages between English, Spanish, Portuguese, and French. Check “Remember my language” to have the tool automatically reopen in your chosen language every time, instead of defaulting back to English. Remembering a language saves the workbook, so if you have inputs on screen the tool asks first: Yes saves the preference and clears the calculator, and No keeps your work and changes the language for this session only.

4.4 Closing the tool

The Close Tool button closes this workbook without saving. If no other workbook is open it also closes Excel; if you have other files open, they are left alone.

5. The 1/3-2/3 Split-Range Valve Calculator

Some jobs need two valves working together instead of one — a small valve that handles the low end of the load precisely, and a larger valve that opens for peak demand. This is called split-range (or 1/3-2/3) valve staging. The OneThirdTwoThird sheet takes the Cv the Calculator sheet just worked out and splits it into a recommended small/large valve pair.

A1									
	A	B	C	D	E	F	G	H	I
1									
2		1/3 - 2/3 VALVE CALCULATOR							
3									
4			1/3rd	2/3rd					
5		Cv 1/3 - 2/3:	7.68	15.36					
6									
7		LB/HR 1/3 - 2/3:	833.3	1,666.7					
8									
9									
10		TURNDOWN ANALYSIS (EFFECTIVE RANGEABILITY)							
11									
12		Valve to Analyze:	Recommended Valve						
13		Curve Type:	Equal %						
14		Rated Rangeability (R):	50:1						
15		Required Cv (design duty):	23.04						
16		Installed Valve Cv:	28						
17		Oversizing Factor (installed / required):	1.22:1						
18									
19		Minimum / Turndown Flow (lb/hr):	50						
20		Flow Basis (design flow, lb/hr):	2,500.0 lb/hr						
21		Required System Turndown (design / min):	50.0:1						
22		Effective Rangeability (R / Oversizing):	41.1:1						
23									
24			SPLIT RECOMMENDED - the analyzed valve's effective rangeability (41.1:1) is less than the required system turndown (50.0:1). A 1/3 - 2/3 valve pair (below) holds control down to the minimum load; a single valve will not.						
25		Recommendation:	Split Valve Pair (1/3 - 2/3):						
26		Small Valve (Low-Load, 1/3):	Globe						
27		Large Valve (Peak, 2/3):	G225S-K (1-in) - Cv 10.0 (target 7.68)						
28			G232S-M (1.25-in) - Cv 20.0 (target 15.36)						
29									

Figure 5 – The 1/3-2/3 Valve Calculator sheet, with a worked split-range example.

5.1 How to use it

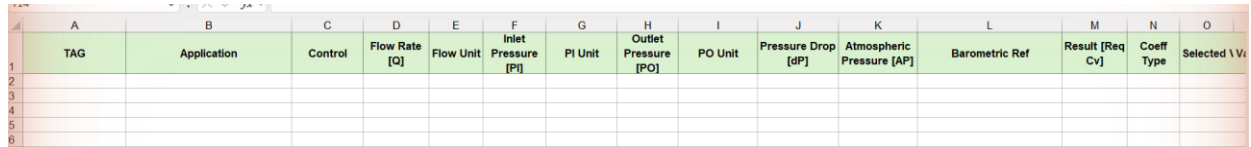
- Size your load on the Calculator sheet first, then pick Recommended Valve or Selected Valve in Valve to Analyze (cell C12) — the analysis stays blank until you do.
- Enter your building's actual Minimum / Turndown Flow (the lowest flow the system realistically needs to deliver, in lb/hr) in the highlighted cell.
- The Turndown Analysis section calculates the effective rangeability needed and tells you, under Recommendation, whether a single valve can handle that turndown or whether a split-range pair is advisable.
- If a split-range pair is recommended, choose Ball or Globe in the Split Valve Pair dropdown to see the small (low-load, 1/3) and large (peak, 2/3) valves as one valve type.

Field / Dropdown	Where it is	Choices
Split Valve Pair (1/3 - 2/3)	OneThirdTwoThird sheet, cell C25	-Select-, Ball, Globe
Why split the valve into two A single valve sized for peak flow often can't control accurately at low flow — it barely has to crack open, and		

small position changes swing the flow a lot. Splitting the job between a small valve (handling roughly the bottom third of the range) and a large valve (handling the top two-thirds) gives you good control across the whole range instead of just at the top end.

6. The Summary Sheet

If you're sizing valves for more than one tag, room, or piece of equipment in the same project, the Summary sheet lets you log each completed calculation as its own row, so you end up with a running record for the whole job instead of re-entering data or losing earlier results.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	TAG	Application	Control	Flow Rate [Q]	Flow Unit	Inlet Pressure [PI]	PI Unit	Outlet Pressure [PO]	PO Unit	Pressure Drop [dP]	Atmospheric Pressure [AP]	Barometric Ref	Result [Req Cv]	Coeff Type	Selected Valve
2															
3															
4															
5															
6															

Figure 6 – Summary sheet column headers — every field captured per logged calculation.

Once you're happy with a result on the Calculator sheet, use the “Add to Summary” action to copy that job's inputs and results (Tag, Application, Control, Flow Rate, Inlet/Outlet Pressure, Pressure Drop, Atmospheric Pressure, the required Cv, Coefficient Type, and the Selected Valve) into a new row here. Over the course of a project this becomes your schedule of valves — useful for a submittal package or for handing off to purchasing.

7. The Tool Box — Extra Engineering Calculators

The Tools sheet is a set of small, independent calculators for everyday HVAC and hydronic math that comes up alongside steam valve sizing — handy even on jobs that aren't primarily about steam.

The Tool Box

Convert Amps to Ohms

Voltage: 24
Amps: 1
Ohms: 48

Convert Watts to Volts

Watts: 5
Amps: 3
Voltage: 2

Convert Watts to Amps

Watts: 5
Voltage: 24
Amps: 0.21

Convert Amps to Watts

Voltage: 24
Amps: 2
Watts: 48

Pump Horse Power

GPM: 10
Total Head: 30
HP: 0.076

Cv = GPM / Sq Root of the Delta P

FLOWRATE: 35.00 GPM
PSI DROP: 4.00 PSI
Cv REQUIRED: 17.50
Kv: 10
= Cv: 11.6

Coil DeltaP Method

Design Flow: 1000.0 GPM
Measured Coil ΔP: 30.0 PSI
Design Coil ΔP: 30.0 PSI
Actual GPM: 1000.00

ΔP Calculator = GPM / Cv Squared

GPM Req'd: 100.00
Cv of the Valve: 240.00
GPM/Cv = 0.42
Adj ΔP = 0.17

Valve Authority

Valve ΔP: 4
Branch ΔP / Coil: 240.00
Valve Authority: 0.50 & Greater is

Valve Cv

Valve Cv: 200

Valve Flow Rated based on DeltaP

ΔP	1	2	3	4	5	6	7	8	9	10	11	12
GPM	200.00	282.84	346.41	400.00	447.21	489.90	529.15	565.69	600.00	632.46	663.32	692.82
ΔP	8.00	565.69										

Fluid Velocity in FPS based on Flow Rate

Pipe Size	1	81.70	115.54	141.51	163.40	182.69	200.12	216.16	231.08	245.10	258.36	270.97	283.02
-----------	---	-------	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------

CvC Calculator

Cv: 200
Valve Size: 2
Pipe Size: 2
Pv: 2
Cvc: 2

By-Pass Valve

Total GPM: 10
Feet of Head: 30
In PSI: 0.076

Figure 7 – The Tool Box sheet — electrical, flow/Cv, and pipe-sizing helper calculators.

7.1 What's included

- Electrical conversions: Amps ↔ Ohms ↔ Watts ↔ Volts, and Pump Horsepower from GPM and Total Head.
- $Cv = GPM / \sqrt{\Delta P}$ and $GPM = K \times \sqrt{\Delta P}$ — the basic hydronic (water-side) valve sizing formulas, for quick liquid-flow Cv checks outside of the steam calculator.
- Coil ΔP method and ΔP Calculator (GPM/Cv squared) — for estimating a coil's actual operating pressure drop from a design condition.
- Valve Flow Rated based on ΔP, and Fluid Velocity in FPS based on Flow Rate — quick reference tables across a range of ΔP and flow values.
- Valve Authority and Branch/Coil pressure-drop ratio calculators — for checking that a control valve has enough authority over its circuit to control well.
- A CvC (Cv Calculator) and By-Pass Valve calculator for auxiliary sizing questions.

These tools are self-contained on this sheet — they don't feed back into the steam Calculator sheet — so you can use them for a quick side calculation without disturbing your steam sizing job.

8. Steam Pipe Capacity Reference (SteamPipeSize Sheet)

This sheet is a reference table, not a calculator you fill in — it's the lookup table the tool itself uses (and that you can use directly) to check how much steam a given pipe size can carry at a given pressure and steam speed.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Capacity (lb/hr)		Pipe Size (inch)														
2	Pressure (psi)	Steam Speed (ft/sec)	0.5-inch	0.75-inch	1-inch	1.25-inch	1.5-inch	2-inch	2.5-inch	3-inch	4-inch	5-inch	6-inch	8-inch	10-inch	12-inch	
3	5	50	12	26	45	70	100	190	280	410	760	1250	1770	3100	5000	7100	
4	5	80	19	45	75	115	170	300	490	710	1250	1800	2700	5200	7600	11000	
5	5	120	29	60	110	175	245	460	700	1000	1800	2900	4000	7500	12000	16500	
6	10	50	15	35	55	88	130	240	365	550	950	1500	2200	3770	6160	8500	
7	10	80	24	52	95	150	210	380	600	900	1500	2400	3300	5900	9700	13000	
8	10	120	35	72	135	210	330	590	850	1250	2200	3400	4800	9000	14400	20500	
9	20	50	21	47	82	123	185	210	520	740	1340	1980	2900	5300	8000	11500	
10	20	80	32	70	120	190	260	520	810	1100	1900	3100	4500	8400	13200	18300	
11	20	120	50	105	190	300	440	840	1250	1720	3100	4850	6750	1300	19800	28000	
12	30	50	26	56	100	160	230	420	650	950	1650	2600	3650	6500	10500	14500	
13	30	80	42	94	155	250	360	655	950	1460	2700	3900	5600	10700	16500	23500	
14	30	120	62	130	240	370	570	990	1550	2100	3950	6100	8700	16000	25000	35000	
15	40	50	32	75	120	190	260	505	790	1100	1900	3100	4200	8200	12800	18000	
16	40	80	51	110	195	300	445	840	1250	1800	3120	4600	6800	13400	20300	28300	

Figure 8 – Steam Pipe Capacity table (lb/hr) by pressure, steam speed, and pipe size.

Find your design steam pressure and target steam speed (ft/sec) in the left-hand columns, then read across to the pipe size column that just meets or exceeds your required flow (in lb/hr) — that's your minimum pipe size. This is exactly the table the Calculator sheet's automatic “min. pipe size” recommendations (Section 3.8) are pulled from, so if you ever want to double check or explain a recommendation to a customer, this is where the numbers come from.

9. Valve Database Reference (Valves Sheet)

This sheet holds the complete Belimo steam valve product data the tool draws on for automatic valve selection and for the manual valve-selection dropdown (Section 3.7.1).

	A	B	C	D	E	F	G	H	I	J	K	L
	Part #	Model	Size	Cv (US)	Kv (m ³ /h)	PI	ValveType	Ball Dia	Ball Area (sq-in)	Seat/Port Area (sq-in)	Curve Type	Rangeability
1	B2050VS-01	VS	0.5	1.00	0.87	35	VS	0.1870	0.0275		Equal %	100
2	B2050VS-02	VS	0.5	2.00	1.73	35	VS	0.2810	0.0620		Equal %	100
3	B2050VS-04	VS	0.5	4.00	3.46	35	VS	0.3750	0.1104		Equal %	100
4	B2050VS-15	VS	0.5	15.00	12.98	35	VS	0.5000	0.1963		Equal %	100
5	B219VS	VS	0.75	30.00	25.95	35	VS	0.6800	0.3632		Equal %	100
6	B220VS	VS	0.75	51.00	44.12	35	VS	0.8600	0.5809		Equal %	100
7	B224VS	VS	1	43.00	37.20	35	VS	0.8700	0.5945		Equal %	100
8	B225VS	VS	1	68.00	58.82	35	VS	1.0000	0.7854		Equal %	100
9	B232VS	VS	1.25	48.00	41.52	35	VS	1.0000	0.7854		Equal %	100
10	B239VS	VS	1.5	84.00	72.66	35	VS	1.2500	1.2272		Equal %	100
11	B240VS	VS	1.5	177.00	153.11	35	VS	1.5000	1.7671		Equal %	100
12	B249VS	VS	2	108.00	93.42	35	VS	1.5000	1.7671		Equal %	100
13	B2050VSS-15	VSS	0.5	15.00	12.98	50	VSS	0.5000	0.1963		Equal %	100
14	B219VSS	VSS	0.75	30.00	25.95	50	VSS	0.6800	0.3632		Equal %	100
15	B224VSS	VSS	1	43.00	37.20	50	VSS	0.8700	0.5945		Equal %	100
16	B249VSS	VSS	2	108.00	93.42	50	VSS	1.5000	1.7671		Equal %	100

Figure 9 – Valve database: part number, model, size, Cv/Kv, pressure rating, ball/port geometry, valve curve, and rangeability for every valve.

9.1 Columns explained

- Part # / Model — the Belimo catalog part number and model family (e.g., VS, VSS, HTCCV).
- Size — valve size in inches.
- Cv (US) / Kv (m³/h) — the valve's rated flow coefficient in both unit systems.
- PI — the valve's maximum rated inlet pressure (used for the pressure-rating check described in Section 3.7.1).
- ValveType, Ball Dia., Seat/Port Area — the physical geometry behind the flow coefficient.
- Curve Type — the valve's inherent flow characteristic (Equal % for most of these models, meaning flow changes proportionally with position — well suited to modulating control).
- Rangeability — the ratio between the valve's maximum and minimum controllable flow; this is what feeds the 1/3-2/3 turndown analysis in Section 5.

You generally won't need to edit this sheet — it updates automatically as new valve models are added to a future release — but it's the place to look if you want to see the raw catalog data behind a recommendation.

10. Standalone Pipe & Valve Velocity Checker (Velocity Sheet)

This sheet is the same velocity-check logic used inside the Calculator sheet (Section 3.8.2), but available as its own standalone tool — useful if you want to check a pipe/valve velocity for a job that was sized outside this tool, or re-check an as-built installation.

A	B	C	D	E	F	G	H
1	Steam Valve Outlet Velocity Check	Valve Type [Select]	Globe		Calculate Outlet Velocity		
2	Steam Quality Override (blank = auto-classify from entropy)	Dry Saturated					
3	Min Pipe Size (SteamPipeSize table) @ 50 / 80 / 120 ft/s	4-in	3-in	2.5-in	Apply Pipe-Size Floor		
4	Steam Flow	P1 Inlet Abs	P2 Corrected Abs	Specific Volume Out (ft ³ /lb)	Steam Classification	Quality (x)	Speed of Sound (ft/s)
5	2493.4 lb/hr	79.70 psia	75.77 psia	5.761	Dry Saturated	1	1621.9
6							
7							
8							
9							
10	Selected Valve Result	Cv	Port Area (in ²)	Area Source	Velocity (ft/s)	Velocity (m/s)	Mach No.
11	Valve (Part#, Size, Type, Cv) B2100VB-024 (1-in, V-Ball, Cv=24.0)	24	0.68	Measured (Ball Area)	845	257.5	0.521
12	RECOMMENDED: B2150VB-055 (1.5-in, Cv=55.0) 91.7 m/s, Mach 0.185 -> Pass						
13	Inlet pipe (2-in Sch 40): 163.3 ft/s (49.8 m/s) - CAUTION (above 40 m/s practical limit, up to 76 m/s may be acceptable for short, insulated, dry runs per some standards)						
14	Outlet pipe (2-in Sch 40): 171.2 ft/s (52.2 m/s) - CAUTION (above 40 m/s practical limit, up to 76 m/s may be acceptable for short, insulated, dry runs per some standards)						
15	Recommendation:						
16	Catalog Scan (filtered by selected Valve Type)						
17	Part#	Size (in)	Cv	Port Area (in ²)	Area Source	Velocity (ft/s)	Velocity (m/s)
18	B2100VB-024	1	24	0.68	Measured (Ball Area)	845	257.5
19	B2150VB-055	1.5	55	1.91	Measured (Ball Area)	300.8	91.7
20	B2200VB-077	2	77	3	Measured (Ball Area)	191.5	58.4
21	B6300VB-207	3	207	5.28	Measured (Ball Area)	108.8	33.2
22	B6400VB-350	4	350	7.19	Measured (Ball Area)	79.9	24.4
23	B6600VB-507	6	507	15.56	Measured (Ball Area)	38.9	11.3
24	B6800VB-865	8	865	25	Measured (Ball Area)	23	7
25	B61000VB-1550	10	1550	44.62	Measured (Ball Area)	12.9	3.9
26	B61200VB-1905	12	1905	64.55	Measured (Ball Area)	8.9	2.7

Figure 10 – The Velocity sheet, showing a completed outlet velocity check with a recommended valve and pipe-size warnings.

Field / Dropdown	Where it is	Choices
Valve Type	Velocity sheet, cell C2	Butterfly, Globe, HTCCV, V-Ball, VS, VSS
Steam Quality Override	Velocity sheet, cell C3	(blank = auto-classify), Dry Saturated, Wet Saturated, Superheated
Target Velocity %	Velocity sheet, cell G4	50, 80, 120
Supply / Outlet Pipe Size	Velocity sheet, cells J2–J3	0.5, 0.75, 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12 (inches)

Enter the steam flow, select a valve type and pipe sizes, and click Calculate Outlet Velocity. The sheet reports velocity in ft/s and m/s and the resulting Mach number for both the selected valve and the inlet/outlet piping, and flags any pipe that's running faster than the practical velocity limit for a short run — exactly as shown in the worked example above, where both the inlet and outlet 2-inch Schedule 40 pipe were flagged as running above the general guideline, with a note that this can still be acceptable for a short pipe run per some engineering standards.

11. The Engineering Behind the Tool, in Plain Language

You don't need to understand the math to use the tool — but if you want to explain a result to a customer, or you're the kind of person who likes to know what's happening under the hood, here's the short version of what the tool is actually doing.

11.1 Why pressure drop (not just flow) sizes the valve

A valve's whole job is to create a controlled restriction. The bigger the pressure drop you allow across the valve (inlet pressure minus outlet pressure), the less restrictive (and therefore smaller) the valve needs to be to pass a given flow. The tool's core formula is a version of the standard valve-sizing equation:

$$C_v = Q \div (1.62 \times \text{inlet pressure, absolute}) - \text{the choked-flow form (see 11.2)}$$

In everyday terms: more flow needs a bigger valve; more inlet pressure lets you use a smaller one. Below the choke point (11.2) the pressure drop across the valve matters as well.

he

11.2 Choked flow — what it means and why the tool warns you about it

Steam (like any compressible gas) can only accelerate through a restriction so fast. Once the pressure drop across the valve gets large enough — specifically, once the outlet pressure drops below roughly 55–58% of the inlet absolute pressure (the exact number depends on the valve) — the steam velocity at the narrowest point (the vena contracta) hits the local speed of sound and cannot go any faster no matter how much you lower the downstream pressure further. This condition is called choked flow, and the pressure ratio where it starts is the critical pressure ratio.

Why this matters to you

Once flow is choked, dropping the downstream pressure further doesn't get you more flow through the same valve — so the tool calculates the valve size using the critical pressure drop, not your full entered pressure drop, whenever it detects choked conditions. If you see a message like “Critical pressure drop reached,” that's the tool telling you it switched to this more conservative calculation to keep the valve sizing accurate, and also flagging that the installation may be noisier or more erosive at that condition — worth a second look, especially on larger pressure drops.

11.3 Valve rangeability and why it matters for turndown

Rangeability is the ratio between the largest and smallest flow a valve can control accurately — a valve with 50:1 rangeability can control flow accurately anywhere from 100% down to 2% of its rated capacity, but below that it starts to lose precision. If your building's actual minimum load is a much smaller fraction of peak than the valve's rangeability allows for, that's when a split-range (1/3-2/3) pair, covered in Section 5, becomes worth considering.

11.4 Equal-percentage valve characteristic

Most of the valves in this product line have an “Equal %” flow characteristic, meaning that equal increments of valve stroke (position) produce equal percentage changes in flow, rather than equal absolute changes. This characteristic pairs well with modulating (throttling) control because it helps keep the loop's overall gain more consistent across the full range of operation — which is part of why “Modulating” control (Section 3.2) generally points toward this valve family.

11.5 Unit conversions, handled for you

Whichever units you enter your flow, pressure, and results in, the tool converts everything to a common internal basis (lb/hr and PSIA) before running the sizing math, then converts the results back to your chosen display units. This is why you can mix, for example, a flow given in MBH with a pressure given in BAR without doing any manual conversion yourself.

12. Troubleshooting & Getting Help

12.1 “Enable Content” bar won't go away / buttons don't do anything

This almost always means macros are still disabled. Close the file, reopen it, and click “Enable Content” on the yellow security bar as soon as it appears (Section 2.1). If your organization's IT policy blocks macros entirely, the tool will not be able to calculate — ask your IT team to allow this specific file, or run it from a trusted location.

12.2 The valve I picked manually was rejected

The tool checks the pressure rating of any manually selected valve (Section 3.7.1) against your entered inlet pressure. If you see a warning message, the valve you picked isn't rated for the job's pressure — choose a different model, or double check that your inlet pressure was entered correctly and in the right unit (PSIG vs. BAR).

12.3 The Recommended Valve says “None suitable – review with TS”

No valve up to 3 times the required Cv both passes the velocity check and is rated for the inlet pressure; the note under the panel says why. Consider two-stage reduction or contact Belimo TS.

12.4 Something else went wrong / unexpected error

If the tool shows an unexpected error message, note what you were entering and what the message said, then use the Report a Bug or Give Feedback button on the Calculator sheet (Section 4.1). The more specific you are, the faster we can track it down.

12.5 Who to contact

For questions this manual doesn't answer, use the Report a Bug or Give Feedback button on the Calculator sheet (Section 4.1), or reach out to your Belimo Regional Application Consultant directly.

Appendix A — All Dropdowns, One Table

A single quick-reference table of every dropdown in the tool, where to find it, and every choice it offers.

Field / Dropdown	Where it is	Choices
Application	Calculator! D6	Coil, Fin Tube, Heat Exchanger, Radiator
Control	Calculator! D8	On/Off, Modulating
Flow Unit	Calculator! D12	BTU/HR, EDR, LB/HR, MBH, KG/h
Inlet Pressure Unit	Calculator! D13	PSIG, BAR
Outlet Pressure Unit	Calculator! D14	PSIG, BAR
Location (Barometric Pressure)	Calculator! D15	Default, Manual / Other, all 50 U.S. states + Canadian provinces (74 total — see table below)
Coefficient Type	Calculator! D28	Cv, Kv
Selected Valve (manual override)	Calculator! J23	-Select-, plus all 59 Belimo steam valve models (see the Valves sheet, Section 9)
Target Velocity (ft/s)	Calculator! C41	50, 80, 120
Velocity Check – Valve Type	Calculator! H42	Butterfly, Globe, HTCCV, V-Ball, VS, VSS
Velocity Check – Steam Type	Calculator! H43	Dry Saturated, Wet Saturated, Superheated
Velocity Check – Pipe/Valve Sizes	Calculator! H44–H46	-Select-, 0.5–12 in. (14 sizes)
View	Calculator! C50	Simple, Detailed
Valve to Analyze / Split Valve Pair	OneThirdTwoThird! C12 / C25	-Select-, Recommended Valve, Selected Valve -Select-, Ball, Globe
Valve Type	Velocity! C2	Butterfly, Globe, HTCCV, V-Ball, VS, VSS
Steam Quality Override	Velocity! C3	(blank = auto), Dry Saturated, Wet Saturated, Superheated
Target Velocity %	Velocity! G4	50, 80, 120
Supply/Outlet Pipe Size	Velocity! J2–J3	0.5–12 in. (14 sizes)

Location / Barometric Pressure — full list

The Location dropdown (Calculator! D15) offers the following 74 entries:

Default • Manual / Other • Alabama • Alaska • Arizona • Arkansas • California • Colorado • Connecticut • Delaware • Florida • Georgia • Hawaii • Idaho • Illinois • Indiana • Iowa • Kansas •

Kentucky • Louisiana • Maine • Maryland • Massachusetts • Michigan • Minnesota • Mississippi • Missouri • Montana • Nebraska • Nevada • New Hampshire • New Jersey • New Mexico • New York • North Carolina • North Dakota • Ohio • Oklahoma • Oregon • Pennsylvania • Rhode Island • South Carolina • South Dakota • Tennessee • Texas • Utah • Vermont • Virginia • Washington • West Virginia • Wisconsin • Wyoming • ...and additional Canadian provinces / territories